Study name:	**Lighthouse Consortium on Obesity Management**
Short name:	**LightCOM**

SOP title:	**Measuring Blood Pressure using a sphygmomanometer (manual or electronic device)**
SOP short title:	**SOP on Blood Pressure**

SOP Version:	**1**	Date:	**09.04.2024**

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TABLE OF CONTENTS

1. PURPOSE

The purpose of this Standard Operating Procedure (SOP) is to describe the standard procedures to be followed when measuring blood pressure from participants during the LightCOM studies which includes LightBAR, LightWAY and LightCARE

2. INTRODUCTION

Measuring blood pressure is a commonly performed procedure to obtain an arterial blood pressure reading from patients, usually to detect the presence of high blood pressure and heart health.

3. SCOPE

This SOP applies to clinical research carried out under the LightCOM research studies which incorporates the individual studies of LightBAR, LightWAY and LightCARE. The sponsor is Department of Medicine, Copenhagen University Hospital Amager and Hvidovre, Copenhagen, Denmark.

4. RESPONSIBILITIES

4.1 Chief Investigator/Principle Investigator or delegate

Will be responsible for ensuring the staff member assigned for obtaining blood pressure measurements are suitably qualified to undertake the procedure and has received adequate and up to date training.

5. SPECIFIC PROCEDURE

5.1 Equipment Check

- Check the specific equipment you are to use for measuring blood pressure and ensure you are familiar with the manufacturer's instruction on use. This will use an electronic device.

Ensure you are using the correct cuff to suit the device you are using and that the cuff is the correct size for the patient, has no signs of wear and tear, and that the fastenings are in place.

To measure the correct cuff size to a participant you must measure the circumference of the patient's mid upper arm with a tape measure in cm. The size of the cuff should between 40% and 80% of the circumference of the arm. It is also recommended to check the manufacturer's instructions for the recommended cuff size to patient arm circumference for the specific equipment you are using.

If the cuff is reusable between participants, clean it fully before and after use, in line with local infection prevention control guidelines.

5.2 Procedure using a Digital Sphygmomanometer

- Check the digital device is in full working order and has been calibrated following the manufacturer's instructions.
- Follow local policy on use of gloves and apron and hand cleaning.
- Allow the participant to rest for a minimum of 5 mins before measuring their blood pressure – this will give a more accurate reading. The participant should remain seated for the measurements
- Check the participant has not:

- Had caffeine, smoked cigarettes, or eaten, and has avoided any exercise at least 30mins before measuring their blood pressure.
- Ensure the patient has an empty bladder as this can affect the blood pressure reading by 10-15mmHg.

Check the participant's arm you intend to take the measurement from and ensure the participant does not have, in the chosen arm:

- An arteriovenous fistula (e.g. for renal dialysis)
- A central or long line in situ or venous cannula
- Lymphoedema
- Presence of a plaster cast, sling or bandage
- Patient has previously had a stroke and has weakness in the arm

- Record the arm that you use in the eCRF.

If the participant is attending a follow up visit, check the previous visit notes in the eCRF as to which arm was used previously and, unless there is good reason not to, use the same arm. If you are using a different arm, record the reason why.

Expose the arm to be used; ensure it is clear of clothing or obstructions; and place the arm, with support, at a height which is level with the patient's mid-sternum.

- Gently palpate the brachial artery to confirm its position.

Place the cuff over the participant's arm covering the brachial artery and secure using fastens as directed by the manufacturer. Ensure there is a width of 2-3cm above the antecubital fossa (two fingers' width) between the patient's arm and the cuff.

If there is an arrow on the cuff, line this up with the brachial artery.

If a participant has a very large arm and the cuff for the electronic device does not fit accurately, revert to using an aneroid sphygmomanometer (see Appendix 1).

Take blood pressure three times in a 5 min window, making sure that the participant's arm is relaxed on a surface to support it.

- Record the readings in the eCRF in the blood pressure section

6. FORMS/TEMPLATES TO BE USED

UK: Use local NHS trust specified Blood Pressure Monitoring Forms
UK and DK: Use the OpenClinica Database as set by the study protocol for recording the participant's blood pressure.

7. INTERNAL AND EXTERNAL REFERENCES

7.1 Internal References

Design and Development of SOP – CRTG Template SOP V3.2

7.2 External References

Measuring Blood Pressure Part 1,2 & Part 3 @2023 Clinical Skills

8. HISTORICAL RECORD OF SOP

Rev# / Date	Written/Revised By	Revision Description	Approval
1. Written	**Odette Dawkins**		
2. Edited			
3.			
4.			
5.			
6.			
7.			
Removed / replaced:			

9. APPENDICES

9.1 Appendix 1: Procedure using an aneroid sphygmomanometer.

- Rapidly inflate the cuff to 20mmHg above the value where the pulse disappears then slowly deflate the cuff completely, using the valve on the bulb. Note the pressure where the pulse reappears – this will give the approximate Systolic Pressure.
- Wait at least 30secs – place the head of the stethoscope lightly over the brachial artery in the antecubital fossa and re-inflate the cuff to 20-30mmHg above the approximate Systolic Pressure.
- Using the valve on the bulb, deflate the cuff at a rate of 2mmHg per second; keep the stethoscope over the brachial artery. Note the pressure reading when you first hear the Korotkoff sounds clearly – this is the Systolic Pressure reading. When the sounds become muffled (K4), this is the first Diastolic Pressure. Note the reading when you hear the last sound – this is the 2nd Diastolic Pressure (K5). Deflate the cuff for a further 10-20mmHg to ensure all sound have disappeared, then deflate the cuff fully.
- Remove the cuff and inform the patient the procedure has been completed.
- Record the patients Blood Pressure using K5 as the Diastolic pressure reading.
- If the patient has a reading over 140/90mmHg it is advised to repeat the blood pressure monitoring twice more – explain to the patient why you are repeating the procedure and record the lower of the last two readings.
- Using the study specific database record the patient ID, date and time and record the blood pressure readings and all other details as deemed by the study eCRF requiremen

9.2 Appendix 2: Procedure using an aneroid sphygmomanometer on a thigh instead of the arm

- If an arm is not possible to take a blood pressure reading, a reading can be taken via a leg cuff.
- To do this you must use the popliteal pulse when auscultating. Note the systolic reading in the thigh is usually 10-40mmHg higher than in the arm but the diastolic pressure will be the same.
- Continue with steps as above for measuring the systolic and diastolic pressure.